Lecture 22.0

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Rearrangement of series:

Ex. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} - \frac{1}{5} + \cdots$

From alternating series test the above series is convergent but it is not an absolutely convergent series as $\sum_{n=1}^{\infty} \frac{1}{n}$ is not convergent.

Let us analyze the following :

	Series	How is it obtained
A	$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$	Original
G	$(1 - \frac{1}{2}) + (\frac{1}{3} - \frac{1}{4}) + (\frac{1}{5} - \frac{1}{6}) + \cdots$	Grouping A
A'	$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots$	Rearrangement of A
G'	$(1 - \frac{1}{2} - \frac{1}{4}) + (\frac{1}{3} - \frac{1}{6} - \frac{1}{8}) + (\frac{1}{5} - \frac{1}{10} - \frac{1}{12}) + \cdots$	Grouping of A'.

Series A : $\frac{1}{1\cdot 2} + \frac{1}{3\cdot 4} + \frac{1}{5\cdot 6} + \cdots > \frac{1}{2}.$

In fact $A = \ln 2 \approx 0.69$

For A':

The bijection for rearrangement :

$f(1) = 1, \quad f(2) = 2, \quad f(3) = 4$

$f(4) = 3, \quad f(5) = 6, \quad f(6) = 8$

$f(7) = 5, \quad f(8) = 10, \quad f(9) = 12$

for $k = 1, 2, 3, \ldots$

$f(3k-2) = 2k-1, \quad f(3k-1) = 4k-2, \quad f(3k) = 4k$

n^{th} block (Each block contains 3 terms)

$$\frac{1}{2n-1} - \frac{1}{4n-2} - \frac{1}{4n}$$

$$= \frac{1}{2(2n-1)} - \frac{1}{4n} = \frac{1}{2}\left[\frac{1}{2n-1} - \frac{1}{2n}\right] > 0$$

Let S'_N be the partial sum for A'.

Then

$$S'_{3n} = \text{sum of first } n \text{ blocks}$$

$$= \sum_{k=1}^{n} \frac{1}{2}\left[\frac{1}{2k-1} - \frac{1}{2k}\right]$$

$$= \frac{1}{2} \sum_{k=1}^{n} \left[\frac{1}{2k-1} - \frac{1}{2k}\right]$$

$$= \frac{1}{2} S_{2n}$$

(S_N = partial sum of A)

(S_{2n} = sum of first n blocks of G)

Given $\lim_{n\to\infty} S_n = \ln 2$

$\Rightarrow \lim_{n\to\infty} S_{2n} = \ln 2.$

$\Rightarrow \lim_{n\to\infty} S_{3n} = \frac{1}{2} \lim_{n\to\infty} S_{2n} = \frac{\ln 2}{2}.$

Then $S_{3n+1} = S_{3n} + \frac{1}{(2n+1)}$

$\Rightarrow \lim_{n\to\infty} S_{3n+1} = \lim_{n\to\infty} S_{3n} = \frac{\ln 2}{2}.$

$S_{3n+2} = S_{3n+1} - \frac{1}{4n+2}$

$\Rightarrow \lim_{n\to\infty} S_{3n+2} = \frac{\ln 2}{2}.$

Thus $A' = \frac{\ln 2}{2}$ while $A = \ln 2.$

Thus rearrangement changes the value of a convergent series.

Lemma: Let $(x_n)_{n\geq 1}$ be a sequence of reals such that $\lim_{n\to\infty} x_{3n+k} = L$ for $k = 0, 1, 2$. Then $\lim_{n\to\infty} x_n = L$.

Proof: Fix $\varepsilon > 0$.

Given $\exists N_1 \geq 1$ s.t. $|x_{3n} - L| \leq \varepsilon \quad \forall n \geq N_1$

$\exists N_2 \geq 1$ s.t. $|x_{3n+1} - L| \leq \varepsilon \quad \forall n \geq N_2$

$\exists N_3 \geq 1$ s.t. $|x_{3n+2} - L| \leq \varepsilon \quad \forall n \geq N_3$

Choose $N = \max\{N_1, N_2, N_3\}, \quad K = 3N+2$

$\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = \varepsilon/3.$

Let $M \geq K$. Apply division algorithm for M.

$M = 3n + r$, where n is some natural number

$r \in \{0, 1, 2\}$

Then $3n = M - r \geq K - 2 = 3N$

$\Rightarrow n \geq N_r \quad r = 0, 1, 2.$

Then $x_M = x_{3n+r} \in r^{th}$ subsequence

$\Rightarrow |x_M - L| \leq \varepsilon$

$$\Rightarrow \boxed{\lim_{n\to\infty} x_n = L}$$

Theorem: Let $\sum_{n=m}^{\infty} a_n$ be an infinite series. Let $f: N \to N$ be a bijection, then $\sum_{n=m}^{\infty} a_{f(n)}$ is absolutely convergent if $\sum_{n=m}^{\infty} a_n$ is absolutely convergent. In fact,

$$\sum_{n=m}^{\infty} a_n = \sum_{n=m}^{\infty} a_{f(n)}.$$

Riemann rearrangement form: A convergent series, which is not absolutely convergent can be rearranged to converge to any real number or made to diverge.

Lecture 22.1

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Theorem (The root test): Let $\sum_{n=m}^{\infty} a_n$ be an infinite series.

Let $\alpha = \limsup_{n\to\infty} |a_n|^{1/n}$. Then

① if $\alpha < 1$ $\sum_{n=m}^{\infty} a_n$ is absolutely convergent.

② if $\alpha > 1$ " " divergent.

③ if $\alpha = 1$ inconclusive.

Proof: Case 1: $\alpha < 1$.

Note that since $|a_n|^{1/n} \geq 0 \Rightarrow \alpha \geq 0$.

i.e. $0 \leq \alpha < 1$.

Then $\exists$ $\varepsilon > 0$ s.t. $0 < \alpha + \varepsilon < 1$.

$$\alpha + \varepsilon > \alpha = \limsup_{n\to\infty} |a_n|^{1/n}$$

$\Rightarrow$ $\exists$ $N \geq m$ s.t. $\forall$ $n \geq N$

$$\alpha + \varepsilon \geq |a_n|^{1/n}$$

$$\Rightarrow \quad |a_n| \leq (\alpha+\varepsilon)^n$$

Since $0 < \alpha + \varepsilon < 1$, $\sum_{n=N}^{\infty} (\alpha+\varepsilon)^n$ is absolutely convergent.

From comparison test $\sum_{n=N}^{\infty} a_n$ is absolutely convergent.

$\Rightarrow$ $\sum_{n=m}^{\infty} a_n$ is also absolutely convergent.

Case 2: $\alpha > 1$.

i.e. $\limsup_{n\to\infty} |a_n|^{1/n} > 1$

$\Rightarrow$ $\forall$ $N \geq m$ $\exists$ $n \geq N$ s.t. $|a_n|^{1/n} > 1$

or $|a_n| > 1$

$\Rightarrow$ $\lim_{n\to\infty} a_n \neq 0$ $\Rightarrow$ $\sum_{n=m}^{\infty} a_n$ is divergent.

Case 3: $\alpha = 1$ (There are both convergent and divergent series with this property).

Remark: If $\lim_{n\to\infty} |a_n|^{1/n}$ exists then $\limsup_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty} |a_n|^{1/n}$.

Claim: Let $(c_n)_{n\geq m}$ be a sequence of positive real numbers. Then

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} c_n^{1/n} \leq \limsup_{n\to\infty} c_n^{1/n} \leq \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}.$$

Proof: $\limsup_{n\to\infty} c_n^{1/n} \leq \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$. $\rightarrow$ ①

Let $L = \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$.

If $L = \infty$, then clearly the inequality ① is correct.

So let L be a finite, nonnegative number (as $\frac{c_{n+1}}{c_n} \geq 0$).

Let $\varepsilon > 0$. Then $L + \varepsilon > L = \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$

$\Rightarrow$ $\exists$ $N \geq m$ s.t. $\forall$ $n \geq N$ $L + \varepsilon \geq \frac{c_{n+1}}{c_n}$

i.e. $c_{n+1} \leq c_n (L+\varepsilon)$

i.e. $c_{N+1} \leq c_N (L+\varepsilon)$, $c_{N+2} \leq c_{N+1}(L+\varepsilon) \leq c_N (L+\varepsilon)^2$.

$$\Rightarrow \quad c_n \leq c_N (L+\varepsilon)^{n-N} \quad \forall\ n \geq N.$$

$$= c_N (L+\varepsilon)^{-N} (L+\varepsilon)^n$$

$$= A (L+\varepsilon)^n, \quad \text{where } A = c_N (L+\varepsilon)^{-N} \neq 0$$

i.e. $c_n^{1/n} \leq A^{1/n} (L+\varepsilon)$

Since $\lim_{n\to\infty} A^{1/n} = 1 \Rightarrow \lim_{n\to\infty} A^{1/n}(L+\varepsilon) = L + \varepsilon$

$$\Rightarrow \quad \limsup_{n\to\infty} c_n^{1/n} \leq (L+\varepsilon) \quad \forall\ \varepsilon > 0$$

$$\Rightarrow \quad \boxed{\limsup_{n\to\infty} c_n^{1/n} \leq L}$$

Similarly, the other direction.

Theorem (Ratio test): Let $\sum_{n=m}^{\infty} a_n$ be a series of nonzero numbers.

Let $\beta = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$, $r = \liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$

① If $\beta < 1$, $\sum_{n=m}^{\infty} a_n$ is absolutely convergent.

② If $r > 1$ $\sum_{n=m}^{\infty} a_n$ is divergent

③ Remaining case Inconclusive.

Proof: Follows from previous claim.